InsightEngine: An End-to-End Data Engineering Platform for Business Intelligence

Team Members

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Overview

Abstract

InsightEngine is an end-to-end data engineering platform designed to collect, ingest, process, transform, store, and analyze data from multiple business sources to support data-driven decision-making. The platform simulates real-world data engineering workflows by generating structured and semi-structured business data using Python and ingesting it through batch and streaming mechanisms. Apache Kafka is used for data ingestion, PostgreSQL is used for operational and analytical storage, and PySpark is used for scalable data processing and transformations. Apache Airflow orchestrates and automates the complete data pipeline, including data validation, cleaning, transformation, and loading into a structured data warehouse based on dimensional modeling principles. The processed data is presented through an interactive Streamlit dashboard to provide meaningful business insights. The project demonstrates key data engineering concepts including source systems, data architecture, storage systems, ingestion strategies, batch processing, data modeling, transformations, and the complete data engineering lifecycle.

Introduction

In today's data-driven environment, organizations generate large volumes of data from multiple sources such as customer transactions, products, payments, inventory, and business operations. Raw data from these sources is often distributed, inconsistent, and difficult to use directly for decision-making. Data engineering provides the processes and technologies required to collect, ingest, store, process, transform, and organize this data into reliable and meaningful information.

InsightEngine is an end-to-end data engineering platform designed to demonstrate the complete data engineering lifecycle. The platform generates realistic business data using Python, supports batch and streaming data ingestion through Python and Apache Kafka, stores operational data in PostgreSQL, and performs data cleaning and transformation using PySpark. Apache Airflow is used to orchestrate and automate the pipeline, while the transformed data is stored in a structured data warehouse using dimensional modeling principles. Finally, an interactive Streamlit dashboard presents business insights. The project integrates key data engineering concepts including source systems, data architecture, storage, ingestion, batch processing, data modeling, transformations, and data quality.

Thank You